SMITKUMAR JAYENDRAKUMAR VELANI

velanismitkumar@gmail.com | +1 857-426-7025 | Portfolio | LinkedIn | GitHub | Boston, MA

Education

Northeastern University, Boston, MA 2026 – 2028
Master of Science in Data Science

Adani Institute of Infrastructure Engineering (GTU) 2021 – 2025
B.E. Information Communication & Technology CGPA: 8.24/10

Work Experience

Unified Mentor Pvt. Ltd. | **Data Analyst Intern** Certificate Jan 2025 – Apr 2025
Stock Market Data Analysis System Link *Python | Pandas | NumPy | Matplotlib | Seaborn | yfinance | Tkinter*

- Integrated **Yahoo Finance API** to fetch real-time **OHLCV** data for **4+ stocks**, automating data cleaning, deduplication and validation using **Pandas** across **250+ daily records**.
- Engineered **11 technical indicators** from scratch (SMA, EMA, Bollinger Bands, RSI, MACD) using fully vectorised **NumPy** and **Pandas** with no external TA library.
- Built an interactive **Tkinter dashboard** with **20 clickable charts** and computed **Sharpe Ratio**, **Max Drawdown** and annualised return metrics for quantitative risk-return profiling.

IBM SkillsBuild | **Data Analytics Intern (CSRBOX)** Certificate Jun 2024 – Aug 2024
Plant Health Prediction System Link *Python | NumPy | Flask | TensorFlow | OpenCV | Keras*

- Deployed a **MobileNetV2 Transfer Learning** CNN achieving **89% diagnostic accuracy** across **15 disease classes** and **3 plant species** (Pepper, Potato, Tomato) on **PlantVillage** dataset.
- Built **OpenCV** preprocessing pipeline (denoising, HSV-based lesion analysis, Canny edge detection) and implemented **two-phase training** with EarlyStopping and ReduceLROnPlateau callbacks.
- Deployed end-to-end **Flask web application** with drag-and-drop interface and real-time CNN inference across all disease classes.

Projects

RAG PDF Chatbot Link
NLP | LLM | RAG Project *Python | LangChain | FAISS | HuggingFace | Gemini API*

- Built an end-to-end **RAG pipeline** using **LangChain** and **FAISS** vector database, processing PDF documents into **500-word overlapping chunks** and storing as **384-dim semantic vectors**.
- Implemented **semantic search** using **HuggingFace sentence-transformers** (all-MiniLM-L6-v2) to retrieve top-3 relevant chunks per query, achieving accurate answers even when user wording differs from document text.
- Integrated **Google Gemini 2.5 Flash API** with custom **prompt engineering** to generate context-grounded answers, tested on 15-page research documents and achieving correct answers on factual queries.

Loan Default Prediction with MLflow Link
MLOps | ML Pipeline Project *Python | XGBoost | MLflow | Scikit-Learn | Pandas*

- Trained 4 ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost) on **32,581 real loan records**, tracked all experiments using **MLflow**; XGBoost achieved best performance with **93.4% accuracy** and **AUC: 0.947**.
- Performed end-to-end data preprocessing including missing value imputation, outlier removal, label encoding and feature scaling on **Kaggle Credit Risk Dataset**.
- Identified **loan grade**, **home ownership** and **loan percent income** as top predictors via XGBoost feature importance and saved best model using **joblib** for deployment.

Multi-Agent Atari Boxing with Deep Q-Learning Link
Reinforcement Learning Project *Python | PyTorch | PettingZoo | OpenCV | Matplotlib*

- Trained two competing **Deep Q-Network agents** in the Atari Boxing environment using **PettingZoo** and **PyTorch** over **200 episodes** on a **T4 GPU**, achieving a **62%/38% win-rate** split.
- Built a **CNN-based DQN** (3 Conv + 2 FC layers) on **4 stacked 84×84 grayscale frames**; implemented **Experience Replay** (10K buffer), **Target Network** and **Epsilon-Greedy** (ε: 1.0 $\rightarrow$ 0.1) from scratch.
- Preprocessed raw **210×160×3 RGB frames** via **OpenCV** (grayscale, resize, normalization) and visualized reward curves and win-rate progression using **Matplotlib**.

Technical Skills

Languages: Python, SQL, JavaScript, PHP, HTML, CSS, SwiftUI
Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch, OpenCV, Flask, LangChain, Bootstrap, D3.js
Tools & Databases: PostgreSQL, MySQL, SQLite, MongoDB, FAISS, MLflow, Power BI, Git, GitHub, VS Code, Jupyter
Machine Learning: CNN, Transfer Learning, DQN, Reinforcement Learning, RAG, LLM Integration, Semantic Search, Vector Embeddings, MobileNetV2